THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The symbol of a switch in the open (OFF) position shows: 1

(a) a cell **(b)** a gap in the circuit
(c) a bulb **(d)** a closed loop

2. The scientific method is best described as: 1

(a) copying a friend's result
(b) a flexible approach that may be repeated and revised
(c) a fixed list of steps always followed in one order
(d) guessing the answer

3. A group of similar cells performing a particular function is a: 1

(a) organ **(b)** organism **(c)** tissue **(d)** organelle

4. A disease that lasts for a long time is called: 1

(a) acute **(b)** viral **(c)** chronic **(d)** sudden

5. A fuse wire has: 1

(a) low melting point **(b)** magnetic property
(c) no resistance **(d)** high melting point

6. Two students get different answers for the same experiment. The most sensible step is to: 1

(a) argue until one gives up
(b) check the method and repeat the experiment
(c) average without checking
(d) ignore both results

7. The function of the cell wall is to: 1

(a) control the cell **(b)** give shape and protection to plant cells
(c) store the genetic material **(d)** make food

8. Night blindness is due to a lack of: 1

(a) vitamin A **(b)** iron **(c)** vitamin D **(d)** iodine

9. If a bulb does not glow in a circuit, a possible reason is: 1

(a) too much light in the room **(b)** a broken connection or a fused filament
(c) the wires are copper **(d)** the cell is too big

10. Which of these must be kept the same in a fair test on puri puffing? 1

(a) temperature being tested **(b)** the result obtained
(c) thickness of the rolled dough **(d)** the conclusion

11. The structure that controls the activities of a cell is the: 1

(a) cell membrane **(b)** nucleus **(c)** vacuole **(d)** cell wall

12. A disease that spreads from an infected person to a healthy person is: 1

(a) communicable **(b)** lifestyle only **(c)** hereditary only **(d)** non-communicable

13. In a closed circuit the current flows: 1

(a) in no direction
(b) from the positive terminal through the circuit to the negative terminal of the cell
(c) from negative to positive only outside
(d) only inside the cell

14. If the results do not support a hypothesis, a scientist should: 1

(a) hide the data **(b)** revise the hypothesis and test again
(c) change the data to fit **(d)** stop science

15. The heart is an example of: 1

(a) an organ **(b)** an organelle **(c)** a cell **(d)** a tissue

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is the nucleus called the control centre of the cell? 2

17. Why are vaccines given to children? 2

18. Why is the filament of a bulb made of tungsten? 2

19. State two safety rules to be followed while working with hot oil or a flame. 2

20. How do bacteria differ from plant or animal cells in terms of the nucleus? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how a fuse protects a circuit. 3

22. What is meant by a fair test? Why does a conclusion from an unfair test cause confusion? 3

23. Differentiate between the cell wall and the cell membrane. 3

24. State three reasons why a person should not take antibiotics on his own. 3

OR

Write three uses of the heating effect of current. 3

25. List three factors that must be kept constant while testing the effect of the thickness of dough on the puffing of a puri. 3

OR

Compare a plant cell and an animal cell by giving three differences. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe how errors can occur while taking measurements and how they can be reduced. 5

OR

Differentiate between bacteria, viruses, fungi and protozoa in a tabular form (any four points). 5

27. How do personal and community hygiene help in staying healthy? Give at least three points for each. 5

OR

Explain the use of electromagnets in cranes and for separating iron. Why is soft iron used? 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND